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Special Issue Article

An Open Source Replication of a Winning Recidivism Prediction Model

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Abstract

We present results of our winning solution to the National Institute of Justice recidivism forecasting challenge. Our team, "MCHawks," placed highly in both terms of accuracy (as measured via the Brier score), as well as the fairness criteria (weighted by differences in false positive rates between White and Black parolees). We used a non-linear machine learning model, XGBoost, although we detail our search of different model specifications, as many different models' predictive performance is very similar. Our solution to balancing false positive rates is trivial; we bias predictions to always be "low risk" so false positive rates for each racial group are zero. We discuss changes to the fairness metric to promote non-trivial solutions. By providing open-source replication materials, it is within the capabilities of others to build just as accurate models without extensive statistical expertise or computational resources.

Keywords

recidivism, machine-learning, prediction, open-science

Introduction

In April 2021 the National Institute of Justice (NIJ) announced a recidivism forecasting challenge. The goal of this challenge was for participating teams to develop a model to predict future recidivism among a sample of persons released from prison to parole. The primary outcome was prediction of re-arrest for any crime at 1-, 2-, and 3-years post-release. A secondary outcome was the development of a "fair

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and accurate" model balancing false-positive predictions between Black and White individuals. We entered this competition in the small-team category under the name "MCHawks" (results can be seen at <https://nij.ojp.gov/funding/recidivism-forecasting-challenge-results>). Our team's prediction model placed in the following categories:

- First Place: Year 2, Male Parolees
- First Place: Year 2, Average Accuracy
- Second Place: Year 2, Female Parolees

We also placed second, third, and fifth in the "Accounting for Racial Bias Category" in the Year 1 and Year 2 categories. As part of this special issue on open methods, we provide our code (all based on open-source libraries in Python and R) to replicate our results, as well as a high-level strategy we used using open-source tools to place in the competition.

It stands to be said at the outset what our article *does not do*, as focusing on predictive modeling has fundamental limitations. Traditional quantitative criminological research almost always focuses entirely on developing hypotheses and using statistical inference to either affirm or falsify those hypotheses. We do not do that here—so although our models include static characteristics like prior criminal history when an individual is released from parole, or the work history of the individual, we ascribe no causal interpretations to these factors. A perpetual critique of observational research is the potential for the effects of such factors to be confounded, but this also means one can build an effective predictive model without all of the factors that cause someone's risk to increase or decrease (Dawes, 1979).

A second major limitation of the work is that we do not make prescriptive judgments about *what one should do with such predictions*. In the discussion we return to this question, in particular how one might structure both incorporating a predictive risk assessment instrument into a community supervision practice, as well as estimate its effectiveness in driving particular outcomes. Building an accurate and fair predictive model does not guarantee that it results in more utilitarian treatment of parolees nor risk mitigation for the communities in which they are released.

The subsequent literature review and methodological sections are thus focused on things we can say based on such a competition—how to build a reasonable predictive model using open-source machine learning libraries. We relegate such further considerations on how such a model is to be used in practice to the final discussion section.

An Overview of Machine Learning for Parole Risk Assessment

Violence risk assessment for use by criminal justice agents has a long history, especially in the case of parole (Harcourt, 2007). For example, Ernest Burgess in conjunction with the Illinois State Board of Parole nearly 100 years ago examined different risk factors (such as whether one was married) was associated with parole failure (Burgess, 1928).

Burgess then tallied up such risk factors into a simple sum score, where those that scored higher were at a much higher risk for subsequent parole failure.

There have been two major advances since the early work on such risk tools. One is via the collection of different measures to understand risk (as well as to understand needs for particular individuals involved with the criminal justice system). For instance, while risk in the original Burgess tool was codified at time of release via static factors (e.g., whether one was married), more recent tools widen the scope of potential measures to dynamic factors such as attitudinal measures (Andrews et al., 2006). A second methodological innovation is the use of more complicated modeling strategies, such as using non-linear machine learning models, to improve predictions (Berk et al., 2009). While these need not be mutually exclusive, the latter is more relevant to the current analysis and the motivating NIJ challenge, as the data maintained by Georgia State are the smaller, limited static risk factors, not much changed from the times of Ernest Burgess.

While simple actuarial tools, like Burgess used, often perform well in criminal justice settings (Dawes, 1979; Wheeler et al., 2019), they do tend to result in a slight loss of accuracy compared to more complicated, machine learning models (Zeng et al., 2017). Although there are certainly some instances in which more traditional, additive regression strategies can perform similarly to non-linear machine learning models such as random forests (Tollenaar & van der Heijden, 2013). Because machine learning models can both learn both simpler additive models and well as more complicated non-linear functions (Berk & Bleich, 2013), one should test both models to see whether one works better on their idiosyncratic data.

So far, we have not been particular about a specific non-linear machine learning model. There are ultimately many; random forests (Berk & Bleich, 2013), boosted models (Ridgeway, 1999), neural-networks, and support vector machines (Ang & Goh, 2013) are a non-exhaustive list of examples. These are in addition to more traditional additive regression models, of which regularized models such as Lasso regression are a part (Tollenaar & van der Heijden, 2013). Our particular approach was to not limit our search to any particular non-linear machine learning model, but to use a test/train set to evaluate the out of sample accuracy for *many* different models (Baćak & Kennedy, 2019; Berk, 2008).

In the end, our internal tests resulted in several different models with very similar accuracy metrics. We ended up using a Lasso logistic regression procedure for Round 1 (which we did not win in the overall results, but scored highly in the fairness category), and an XGBoost model for Round 2 (which we scored highly for all measures). We refer to Chen and Guestrin (2016) for an overview of the XGBoost open-source machine library we used, but in short it is an ensemble learner, where one has a base model (here decision trees), and then subsequent models are fit on the prior model's residuals, attempting to improve upon the prior model's mistakes. One then combines those decision models together, and repeats the process many times. The intuition is that it is better to combine many weak models together than it is to rely on a singular overall model. In many practical applications these types of models tend to perform quite well, however they come at the cost of interpretability (Molnar, 2020). Boosted

models can generate predictions that are accurate on new data, but it may not be clear what variable(s) lead to an individual prediction. This is in contrast to more conventional additive models (such as linear regression) where coefficient values have a more immediately clear interpretation. In any case, any choice of model results in a tradeoff of performance and interpretability, which we discuss below.

Methods Overview

Data and Variables

The data used for the recidivism challenge comes from the State of Georgia on 25,385 individuals released to parole in the years 2013, 2014, and 2015. For privacy reasons, the data is limited to only those White and Black individuals. The outcome variable, recidivating, is measured as a new *arrest* (for either a felony or misdemeanor) within the 3 years follow up. This is subsequently broken down for the competition intro predictive recidivism within Year 1, within Year 2, or within Year 3.

The variables available for prediction in all rounds are: gender (male/female), race (Black/White), age at release (in several age bins), residence geographic public use micro area, gang affiliation, supervision risk score (1–10, 1 = lowest), supervision level (standard, high, specialized), education level, number of dependents at prison entry, prison offense (violent non-sex, violent-sex, property, drug, other), prison years, prior felony arrests, prior misdemeanor arrests, prior violent arrests, prior property arrests, prior drug arrests, prior probation/parole violations, prior domestic violence charges, prior gun charges, prior felony convictions, prior misdemeanor convictions, prior violent convictions, prior property convictions, prior drug convictions, prior domestic violence convictions, prior gun convictions, prior parole revocations, prior probation revocations, condition of mental health or substance abuse programming, condition of cognitive skills or education programming, and condition of no victim contact or electronic monitoring or restitution or sex offender registration. There are additional variables available at later rounds for forecasting based on prior supervision related activities; violations for electronic monitoring, any violation for not following release instructions, violations for failure to report, violations for moving without permission, number of parole delinquency reports, number of program attendances, number of program unexcused absences, number of residence changes during parole, average days on parole between drug tests (continuous), percent drug tests positive for marijuana (continuous), percent drug tests positive for cocaine (continuous), percent drug tests positive for methamphetamine (continuous), percent drug tests positive for other substances (continuous), percent days employed while on parole (continuous), jobs per year on parole (continuous), and employment is not required.

Unless noted as continuous in the above paragraph, all variables in our modelling strategies were recoded into 0/1 dummy variables, with one referent category omitted. This included a category for missing data, so no missing data imputation was necessary. (The continuous variables contained no missing information for the relevant rounds.) Although we attempted to ordinally encode some of the variables and tested

Table 1. Selected Descriptive Statistics for Training and Test Samples.

Variable	Train sample	Test sample
Gender		
Female	2,217	950
Male	15,811	6,857
Race		
Black	10,313	4,534
White	7,715	3,273
Age at release		
18–22	1,454	612
23–27	3,611	1,565
28–32	3,449	1,533
33–37	2,975	1,296
38–42	2,040	953
43–47	1,858	762
48 or older	2,641	1,086
Education		
At least some college	3,198	1,407
High school diploma	7,946	3,444
Less than HS diploma	6,884	2,956
Prison offense		
Drug	3,617	1,573
Other	1,913	866
Property	5,797	2,487
Violent/non-sex	3,797	1,678
Violent/sex	583	247
Prior arrest episodes violent		
0	7,705	3,344
1	5,083	2,204
2	2,503	1,110
3 or more	2,737	1,149
Recidivism arrest Year 1		
False	12,651	5,460
True	5,377	2,347
Recidivism arrest Year 2		
False	14,775	6,493
True	3,253	1,314
Total *N*	18,028	7,807

models (e.g. coded education as less than high school = 0, high school = 1, some college = 2, etc.), the dummy variable approach produced better forecasts in our internal tests.

Table 1 displays a selection of, not of all variables (simply for space, those interested in more details can download the data themselves and peruse), broken down by the

training and test datasets defined by NIJ. The data and the description of that data available for participants is available at <https://data.ojp.usdoj.gov/stories/s/daxx-hznc>.

Scoring Metrics

NIJ defined two different scoring metrics to which the competition would be judged. The first is the Brier Score (BS), defined as:

$$\mathrm{BS} = 1/n \cdot \sum_{i}^{n} \left(\hat{P}_i - y_i \right)^2$$

Where $\hat{p}_i$ is the forecasted probability for individual i, and y_i is the observed outcome of the individual (0 for not recidivating and 1 for recidivating in that particular wave), and n is the total sample size. The Brier score is the average of squared differences, which is more typically used for continuous regression predictions, although is still applicable to measuring the accuracy of predicting binary outcomes.

The Brier score is then judged for males, females, and an *average* of the BS for the male and female subset of data (not the pooled data BS). These metrics are then applied only to the (unobserved to competitors at the time), hold out testing sample. Lower values of the BS mean the overall predictions are more accurate.

The second metric which NIJ devised is a fairness score (FS), intended to encourage solutions that also minimize differences in false positive (FP) rates between White and Black parolees. False positives occur when a parolee *does not recidivate*, and they subsequently go on to not recidivate in this sample. So, for example, if there were 100 parolees who did not recidivate, and 30 of them were predicted to be high risk, the false positive rate for that sample would be 0.3. The FS is defined as:

$$\mathrm{FP}_{\mathrm{diff}} = 1 - \left| \mathrm{FP}_{\mathrm{White}} - \mathrm{FP}_{\mathrm{Black}} \right|$$
$$\mathrm{FS} = \left(1 - \mathrm{BS}\right) \times \mathrm{FP}_{\mathrm{diff}}$$

Where $\mathrm{FP}_{\mathrm{diff}}$ is one minus the absolute difference in the false positive rate between White and Black parolees. FS is then a weighted metric combining both the BS and the FP metrics together via multiplication. For example, if one had a BS of 0.2, and the $\mathrm{FP}_{\mathrm{diff}}$ was 0.8, the weighted metric would be 0.64. To determine whether a prediction is classified as high risk, a threshold of 0.5 is used by NIJ.

The competition then broke down these metrics for each wave, recidivating and Years 1, 2, and 3, as well as between males, females, and a pooled category for the Brier score. For the Brier scores, the competition was also broken down into small teams (which MCHawks competed in), versus large teams. The fairness metric was not broken down between small and large teams. The metrics, as well as competition rules, can be seen at <https://nij.ojp.gov/funding/recidivism-forecasting-challenge>.

Analysis Methods

For an overview of the methods, we used to determine models, at the time of the competition only the training data was available, not the test data (which are only available

after the competition is finished). Subsequently we used an internal cross validation approach to evaluate a series of different models for both the BS and FS metric.

Models we evaluated encompassed: logistic regression, lasso penalized logistic regression, XGBoost, random forests, and a custom deep learning model (similar in architecture to logistic regression, although the loss function to train the parameters was either the Brier score or the fairness metric). We also evaluated whether individual models separating out males and females would result in better predictions (it did not), and so all models we used were based on the pooled male/female samples, but included a dummy variable indicating the sex of the parolee.

Our winning model for Round 2 was fit using a gradient boosted decision tree employing XGBoost (Chen & Guestrin, 2016). XGBoost is a widely used algorithm for fitting gradient boosted models (GBM) and has a strong track record in various prediction competitions. Prior to fitting the model, we first expanded the full training dataset from NIJ into a 18028 x 135 data matrix. For the Year 2 predictions we filtered out individuals who were arrested at Time 1, leaving us with 12,651 observations. No additional variables were added or constructed from the existing data. The first step in fitting a GBM is the tuning of the model parameters. This involves adjusting key parameters in a way to minimize overfitting (learning too much from the training data) and underfitting (learning too little from the training data). In more general terms, this step reflects a strategy of balancing bias and variance (James, et al., 2013). We employed a grid search strategy, where we iteratively fit models using different combinations of the major XGBoost parameters and calculated an estimate of out-of-sample performance via five-fold cross validation. The most important parameters tuned were:

- max_depth: 6
- eta: 0.005
- min_child_weight: 5
- max_delta_step: 5
- subsample: .75

max_depth and min_child_weight control the overall model complexity, eta controls the stepsize per iteration, and max_delta_step and subsample help control overfitting. Rather than attempting to optimize the Brier score directly, we used area under the curve (AUC) to determine the optimally fitting model. In internal tests, we found models optimized to AUC had slightly higher out of sample Brier scores than XGBoost models trained directly to optimize Brier scores, although differences are admittedly minor. (We also chose an XGBoost model for Round 2, since Round 1 we chose a L1 regularized logistic regression, and ultimately out of sample predictions were near equivalent for the two models.)

We used a potential of 2000 fitting iterations for the XGBoost models, but also had an early stopping number of 20 rounds (if the AUC did not improve in 20 straight iterations, the algorithm would stop.) The AUC value for this model was approximately .74. We then used these hyperparameters found during the cross-validation phase, and then retrained the model on the full training dataset available.

We have provided our code at <https://github.com/gmcirco/nij_mchawks>, while not as organized as a final project, one can see the different local experiments we did (in python and R code) in determining our final models we used for the competition.

For each round, we bias our predictions to never broach the threshold of 0.5—this ensures that the fairness metric, the difference in false positive rates between the two groups, is always 0. So, for example, if the predictive model gave a predicted probability of 0.62, we simply clipped the probability to be 0.4999 for the competition.

Similar to Mohler and Porter (2021), we found that given the volatility in the fairness metric, simply biasing predictions to never be above 0.5 resulted in the best performance on this metric. This is even better than fitting a model to minimize this loss function in sample, because the volatility of out of sample can be potentially quite large. This biasing is obviously a trivial solution (and largely defeats the purpose of risk assessment), and so we provide an alternative discussion of a potential modification of the fairness function that is less likely to result in the trivial "predict everyone low risk".

We note that ultimately one can subsequently take the data provided and no doubt concoct different models that are slight improvements on the competition metrics (and indeed several teams had *slightly* better metrics than ours). The benefit of the competition format however is there is no way for competitors to overfit to the withheld testing dataset. Thus, the external validity of the measures on the test dataset are ensured via the competition.

Results

Table 2 displays our results for Rounds 1 and 2 (we did not place in any category for Round 3), with notes on placement levels where applicable. For Round 1, our lasso logistic regression did not place in the main Brier score competition for small teams. Had we not biased our predicted probabilities to always be below 0.5, we would have placed either third or fourth in the small team category for all three Brier score competitions. Biasing the predictions in such a way only resulted in a Brier Score reduction of 0.001 or 0.002 in each of the categories, but such a small difference was enough to push one out of the leaderboard. Generally for all leaderboards rankings were only distinguishable in the thousands place for both the Brier Score and fairness metrics.

In Round 1 for the fairness metric, our team placed third in the Male parolees and second in the female parolees' categories. The winning solution for Female parolees, team PASDA, in this round used a very similar model to the one we described, along with biasing predictions is a similar way (Mohler & Porter, 2021). Because we biased predictions to always be below 0.5, the false positive rate for each sex group was 0/0, which for the competition is defined as 0. As such the fairness score is simple one minus the Brier Score. (False negative rates are then simply the total proportion of the sample recidivating, since we predict no one to recidivate using the 0.5 threshold approach.)

For Round 2, we had the model with the highest Brier Score in the small team category for males (0.1658) and average accuracy (0.1450), and second place for females

Table 2. Metrics and Team Rankings for Rounds 1 and 2.

	Brier score (team ranking)	Fairness metric (team ranking)
Round 1		
Male	0.1931 (>4)	0.8069 (3)
Female	0.1558 (>4)	0.8442 (2)
Average	0.1745 (>4)	X
Round 2		
Male	0.1658 (1)	0.8342 (5)
Female	0.1242 (2)	0.8758 (2)
Average	0.1450 (1)	X

Table 3. Comparison of Brier Scores for Round 2 Among Small Teams.

Small team	Brier score	Rank
Male results		
MCHawks	**0.1658**	**1**
Oracle	0.1670	2
CategOracles	0.1679	3
VT-ISE	0.1685	4
Female results		
Oracle	0.1233	1
MCHawks	**0.1242**	**2**
VT-ISE	0.1260	3
DEAP	0.1263	4

Results for other teams available at https://nij.ojp.gov/funding/recidivism-forecasting-challenge-results.

(0.1242). This is despite biasing the results downward to always be below 0.5 in the predictions. Table 3 lists the comparison of our teams Brier Scores (in bold) in comparison to the rest of the leaderboard. Among small teams, less than 0.003 separates the top from the bottom leaderboard scores. (The best large team score for this round was 0.1542 and 0.1196 for Males/Females respectively.) Differences in the fairness competition for Round 2 (which includes both small and large teams), we placed fourth in the Male parolee's category, and second in the female parolee's category.

In general, for the competition the larger teams did outperform the smaller teams for all the metrics considered, but the differences in such metrics were very small overall. A supplementary analysis by Wheeler (2020a) suggests that the standard deviation of the Brier score was just under 0.002 using a simulation approach with varying train/test splits. Thus a team's performance who was not even in the top 5 could potentially be the top performing model if a different train/test split had been used.

Although not used for the competition, we also evaluated the area under the curve (AUC) metric for models. Generally, those with lower Brier scores also had better

AUC statistics. It was also the case that the AUC metric appeared to have a higher variance than the Brier score metric as well. Our models tended to have an AUC for this data around 0.75.

Although not directly incorporated into the modelling strategy, one particular aspect of the fairness metric we noticed was that the *expected* false positive rate for the racial groups over the 0.5 threshold was very similar (also noted by Mohler and Porter, 2021). For example, if one has a set of predicted probabilities over 0.5 for a sample $\{0.6, 0.7, 0.8\}$, the expected false positive rate among those three cases would be $1 - \overline{p}$ (the mean average of the probability for those predicted cases over the threshold); in this hypothetical an *estimated* false positive rate of 30%. Calculating this for both males and females, most models we tested had *very* similar expected false positive rates between racial groups, and this suggests why some of the more accurate Brier Score models also scored well in the fairness category—more accurate models (here) tended to also be quite fair according to this metric.

Finally, we describe the feature importance for the resulting models. Although to be clear, the approach we took was largely atheoretical—we encoded all of the features as binary dummy variables. Additionally, one should be aware that these factors are not to be interpreted in a causal fashion and can be potentially misleading reduced form summaries of the very non-linear XGBoost predicted model used (Rudin, 2019).

While the final model was fit on all 135 features, only a small subset of these features were especially useful for the final prediction. Because the model used does not rely on the conventional frequentist framework, we cannot evaluate whether any individual predictor is statistically "significant." However, we can evaluate the individual contribution to the final prediction for each feature. To accomplish this, we evaluated the gain and frequency for each feature. Gain represents the proportional contribution of a feature to the overall model, based on the reduction in overall error for each of the feature's splits. Frequency is the percentage of times a feature is present in a tree (Molnar, et. al., 2018). Together, these indicate which features are most important to the final prediction. Below, Figure 1 displays the gain (blue) and frequency (orange) of the top 20 predictors in our model arranged by gain.

Among these, the five most important features were: (1) the number of jobs, per-year, while on parole; (2) the percentage of days employed while on parole; (3) the average number of days on parole between drug tests; (4) their supervision risk score; and (5) the percentage of drug tests positive for THC/marijuana. Interestingly, race (White vs. Black) is not among even the top 20 predictors in the model. Based on this, the first feature might represent an irregular work history—for example, an individual who cannot hold a steady job—while the second feature reflects largely a more consistent work history.

Discussion

To recap the main contributions of the paper, we have described the modelling strategy used in our winning solution to the NIJ recidivism forecasting challenge. In providing open-source replication code, we have provided the means for anyone with access to

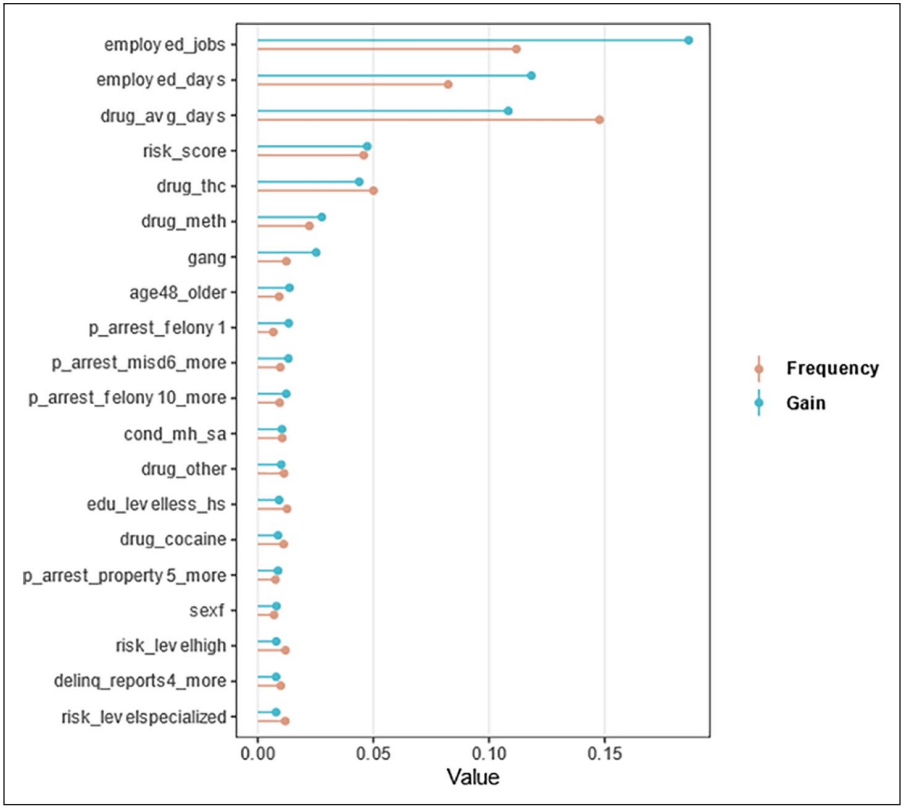

Figure 1. Variable importance for factors in the XGBoost model.

typical computing resources to be able to replicate such results. For interested criminal justice agencies wanting to implement their own predictive models using their own local data, it is certainly within a capable analyst's ability to leverage similar predictive models for their own internal processes.

Although the models we provided were not the best (in various categories the large teams outperformed our models), in practice the variance of the metrics used to score the competition are larger than the differences in the rankings (Wheeler, 2020a). As such, our models performed *nearly* as well as all the winning solutions. Thus, we believe the greatest contribution of the article is in and of itself the open-source code illustrating our modelling strategy. It does not take great statistical expertise, nor special computing requirements, to replicate our findings.

This article then provides a blue-print for others wishing to develop machine learning risk predictions using their own historical data. It is likely the case that an instrument developed on Georgia data will not perform as well for other jurisdictions (nor that other jurisdictions may have access to the same information). Thus, the ability for

analysts to develop models locally is a critical step to applying predictions in actual practice for criminal justice agencies. In terms of future research applications, we discuss two high level goals: first is amending the particular fairness metric to not bias the predictions to a trivial solution, and second how one might incorporate such predictions into practice and evaluate the performance of such a risk assessment instrument.

In terms of future considerations for the fairness metric, there are two obvious issues with the current metric as proposed by NIJ. First, the threshold of 0.5 to categorize an individual as high risk is likely misinformed. One would need to conduct a cost-benefit analysis on how such predictions will be used by criminal justice agents to determine what an appropriate threshold would be. For instance, if these are used to assign more intensive supervision for parolees, there is a limit on how many individuals can be assigned high risk. One may then take the threshold that approximately fills up this queue based on historical values.

Another approach would be to conduct a cost-benefit analysis on a case-by-case basis. The cost of assigning more intensive supervision (both in terms of labor costs for the agency, as well as for costs of the additional government oversight to the parolee) should be balanced with the estimated benefits of that assignment (e.g., reduced recidivism) (Stevenson & Mayson, 2021). This may produce a threshold either much higher or much lower than simply a probability of 0.5. Given the difficulty of conducting such a cost benefit analysis, future competitions may estimate such metrics at different thresholds and average overall results together.

A second consideration is the fairness metric considered itself. The multiplicative term results in large variances, whereas historical fairness penalties to model terms tend to be additive (Mohler & Porter, 2021). So here a slight change to the metric we believe would be more appropriate in practice:

$$\mathrm{BS} + \lambda \times \mathrm{FP}_{\mathrm{diff}}$$

Where lambda is a term to adjust how much one penalizes the false penalty term relative to the Brier Score. Given the typical differences among solutions on the leaderboard for the competition, a lambda value of 0.1 could still result in meaningful shifts among winning solutions but result in much less variance in the overall score due to small sample differences in the FP rate between groups. For a criminal justice agency considering the benefit of balancing false positives, they may then also draw a trade-off curve to determine how much accuracy one is willing to sacrifice to approximately balance the in-equitable false positive rates between groups (Wheeler, 2020). Defining fairness in practice will ultimately involve both empirical utility estimates as well as ethical considerations.

We believe an important future research goal is determining metrics to not only build such models, but to *monitor* those models as used in practice. While minor deviations from false positive rates in small samples may not signify problems, where to draw the line in practice and how to determine if equitable outcomes are being achieved in real life, noisy data are more difficult. While one can build models that penalize differences in the false positive rates in sample (Mohler & Porter, 2021; Short & Mohler, 2020), we believe an incremental model evaluating such divergences over

time and adjusting how one assigns risk status is likely to be an overall more effective strategy for ensuring fairness (Chohlas-Wood et al., 2021; Kleinberg et al., 2018).

We additionally do not broach the question of whether constraining false positive rates between groups makes the most sense as a metric of fairness. There are ultimately many different metrics one may consider (Berk, 2016). One thing to note is that if one wishes to apply similar metrics to balance false negative rates, that can be accomplished by simply switching outcomes (predicting no recidivism), and applying the same metrics or modelling strategies. In the end though there are ultimately trade-offs, one may have to prioritize particular fairness metrics while making other fairness metrics more inequitable (Corbett-Davies et al., 2017).

While balancing false positives has been popularized beginning with a public critique of the COMPAS algorithm by *ProPublica* (Angwin et al., 2016), in some risk assessment criminal justice applications assessing false positives in practice is not even viable. For example, if one uses a predictive instrument to hold someone pretrial, one does not observe any on-policy feedback for false positives (holding pre-trial and no recidivism), only false negatives (releasing into the community and committing a new offense before the trial). One will need to tailor the definition of fairness to the specific application and sample—we believe it is unlikely for there to be a one size fits all approach.

The last aspect of future research we end our discussion on is *how to incorporate such predictions into practice*. Even if one develops a predictive algorithm, and it is reasonably accurate on historical example, those facts go not guarantee the risk assessment will translate into more utilitarian outcomes in practice (Stevenson & Doleac, 2021). This involves two aspects: 1), how are the predictions used in practice, 2) how does one evaluate the efficacy of those decisions.

When one considers assessment for high risk in the case of parolees, two immediate uses are the condition of parole itself are informed by the risk assessment, or perhaps more commonly whether a parolee is tagged for more intensive supervision (Duwe & McNeeley, 2021). It is also possible to amend predictive algorithms to assign individuals to more supportive reentry social services as well (Hamilton et al., 2021). While historically such predictive instruments are often used as guidelines in an additional structured risk assessment (so a human ultimately makes a decision), these have historically performed quite poorly, both in terms of accuracy (Meehl, 1954), as well as mitigating racial biases (Stevenson & Doleac, 2021). Thus, we believe a better system is to automate such decisions as being placed on high-risk parole, and subsequently create external facing instruments to ensure the effectiveness and fairness of such automated decisions (Morgan et al., 2017).

A second aspect involves evaluating whether such a decision-making tool effectively accomplishes the goals it sets out to do—to increase public safety, promote efficiency in the criminal justice system, and balance equitable outcomes among cognizable racial groups. While not easy, there are two ways in which such evaluations can be done in a rigorous manner. One, is that there is an element of randomness in who is assigned the more intensive outcome (Harcourt, 2007). One can then use this randomness to evaluate whether the assigned treatment is effective in reducing recidivism (Angrist & Pischke, 2008). While we find this is unlikely to be an appealing

alternative in practice due to ethical concerns, one situation this is likely to occur in is via resource constraints. In the case of intensive supervision or allocating social services, it may be the case that there is not enough capacity to serve the entire high-risk population. In that case, random assignment seems a reasonable solution, which has the added benefit of allowing one to measure the effectiveness of those treatments in situ over time.

In the case in which random assignment is not feasible, an alternative that still provides a strong, counterfactual estimate of the efficacy of treatment assignment for a risk prediction instrument is a regression discontinuity design (Berk & Rauma, 1983; Rhodes & Jalbert, 2013). In this design, one fits an underlying model predicting recidivism based on the initial assigned predicted risk score, and then estimates a bump in the underlying function at the threshold for particular treatment assignment. While this approach requires more unverifiable assumptions than the random assignment approach, such as the underlying functional form for estimating the counter-factual effectiveness of the intervention, it is still more rigorous than simply evaluating recidivism overall for different risk groups. In practice it may be those assigned high risk and more intensive supervision still recidivate at higher rates than those predicted low risk, but recidivate *less* than they would absent the assigned treatment.

Ultimately one needs not only demonstrate a predictive algorithm is reasonably accurate in historical samples, but also demonstrate that it is effective in amending real life decisions and producing better outcomes. More accurate predictive outcomes may be necessary, but are not sufficient by themselves to accomplish that goal.

Authors' Note

Data and code to replicate the findings can be obtained from https://github.com/gmcirco/nij_mchawks

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